

Finding the Adjugate (Adjoint) of a Matrix

1. Introduction to Adjugate of a Matrix

The **adjugate** (also called the **classical adjoint** in many textbooks) of a square matrix is the transpose of its cofactor matrix.

It is written as:

$$\text{adj}(A)$$

The formula is:

$$\text{adj}(A) = (\text{Cofactor Matrix})^T$$

The adjugate is an important step because it is used to find the inverse of a matrix:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

2. Steps to Find the Adjugate of a Matrix

To find the adjugate of a matrix:

Step 1:

Find the **minor** of every element.

Step 2:

Convert every minor into a **cofactor** using:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Step 3:

Arrange all cofactors into the **cofactor matrix**.

Step 4:

Transpose the cofactor matrix.

The result is the **adjugate matrix**.

3. Adjugate of a 2×2 Matrix

Consider:

$$A = \begin{bmatrix} a & b & c & d \end{bmatrix}$$

Step 1: Find the Minors

Minor of a

Remove row 1 and column 1:

$$M_{11} = d$$

Minor of b

Remove row 1 and column 2:

$$M_{12} = c$$

Minor of c

Remove row 2 and column 1:

$$M_{21} = b$$

Minor of d

Remove row 2 and column 2:

$$M_{22} = a$$

The minor matrix is:

$$M = \begin{bmatrix} d & c & b & a \end{bmatrix}$$

4. Step 2: Find the Cofactor Matrix

Use the sign pattern:

$$\begin{bmatrix} + & - & - & + \end{bmatrix}$$

Applying signs:

$$C = \begin{bmatrix} +d & -c & -b & +a \end{bmatrix}$$

This is the cofactor matrix.

5. Step 3: Transpose the Cofactor Matrix

The adjugate is:

$$\text{adj}(A) = C^T$$

Therefore:

$$\boxed{\text{adj}(A) = \begin{bmatrix} d & -b & -c & a \end{bmatrix}}$$

This formula can be directly used for any 2×2 matrix.

6. Example: Finding Adjugate of a 2×2 Matrix

Given:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \end{bmatrix}$$

Step 1: Find Minors

The minors are:

For element 2:

$$M_{11} = 5$$

For element 3:

$$M_{12} = 4$$

For element 4:

$$M_{21} = 3$$

For element 5:

$$M_{22} = 2$$

Minor matrix:

$$M = \begin{bmatrix} 5 & 4 & 3 & 2 \end{bmatrix}$$

Step 2: Find Cofactor Matrix

Apply signs:

$$C = \begin{bmatrix} + & - & - & + \end{bmatrix}$$

Therefore:

$$C = \begin{bmatrix} 5 & -4 & -3 & 2 \end{bmatrix}$$

Step 3: Transpose the Cofactor Matrix

Transpose:

$$\text{adj}(A) = C^T$$

Interchange rows and columns:

$$\boxed{\text{adj}(A) = \begin{bmatrix} 5 & -3 & -4 & 2 \end{bmatrix}}$$

7. Adjugate of a 3×3 Matrix

Consider:

$$A = \begin{bmatrix} a & b & c & d & e & f & g & h & i \end{bmatrix}$$

The cofactor matrix is:

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{21} & C_{22} & C_{23} & C_{31} & C_{32} & C_{33} \end{bmatrix}$$

The adjugate is:

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & C_{31} & C_{12} & C_{22} & C_{32} & C_{13} & C_{23} & C_{33} \end{bmatrix}$$

Notice:

- Rows become columns.
- Columns become rows.

8. Example: Finding Adjugate of a 3×3 Matrix

Given:

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 4 & 5 & 1 & 0 & 6 \end{bmatrix}$$

We first find all cofactors.

Cofactor C_{11}

Remove row 1 and column 1:

$$C_{11} = \begin{vmatrix} 4 & 5 & 0 & 6 \end{vmatrix}$$

Calculate:

$$C_{11} = 4(6) - 5(0)$$

$$C_{11} = 24$$

Cofactor C_{12}

Remove row 1 and column 2:

$$C_{12} = \begin{vmatrix} 0 & 5 & 1 & 6 \end{vmatrix}$$

Calculate the minor:

$$(0)(6) - (5)(1) \\ = -5$$

Apply negative sign:

$$C_{12} = 5$$

Cofactor C_{13}

Remove row 1 and column 3:

$$C_{13} = \begin{vmatrix} 0 & 4 & 1 & 0 \end{vmatrix}$$

Calculate:

$$(0)(0) - (4)(1) \\ C_{13} = -4$$

The first row of the cofactor matrix is:

$$\begin{bmatrix} 24 & 5 & -4 \end{bmatrix}$$

The same process is repeated for the remaining six cofactors.

The complete cofactor matrix is:

$$C = \begin{bmatrix} 24 & 5 & -4 & -12 & 3 & 2 & -2 & -5 & 4 \end{bmatrix}$$

9. Finding the Adjugate

Transpose the cofactor matrix:

$$\text{adj}(A) = C^T$$

Therefore:

$$\boxed{\text{adj}(A) = \begin{bmatrix} 24 & -12 & -2 & 5 & 3 & -5 & -4 & 2 & 4 \end{bmatrix}}$$

10. Relationship Between Determinant and Adjugate

A very important identity is:

$$A \times \text{adj}(A) = \det(A)I$$

where:

- A = original matrix
- $\text{adj}(A)$ = adjugate matrix
- $\det(A)$ = determinant
- I = identity matrix

This identity is the reason the inverse formula works.

11. Using Adjugate to Find Inverse

The inverse formula is:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

Example:

For:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \end{bmatrix}$$

We already found:

$$\text{adj}(A) = \begin{bmatrix} 5 & -3 & -4 & 2 \end{bmatrix}$$

The determinant is:

$$\begin{aligned} \det(A) &= (2)(5) - (3)(4) \\ &= 10 - 12 \\ &= -2 \end{aligned}$$

Therefore:

$$A^{-1} ===== \frac{1}{-2} \begin{bmatrix} 5 & -3 & -4 & 2 \end{bmatrix}$$

12. Important Points to Remember

1. Adjugate is **not** the same as the inverse.

Wrong:

$$\text{adj}(A) = A^{-1}$$

Correct:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

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2. The adjugate exists for every square matrix.

3. The inverse exists only when:

$$\det(A) \neq 0$$

4. Finding adjugate requires:

Minors $\rightarrow$ Cofactors $\rightarrow$ Transpose

Summary Table

Step	Operation	Result
1	Find minors	Minor matrix
2	Apply signs	Cofactor matrix
3	Transpose cofactors	Adjugate matrix
4	Divide by determinant	Inverse matrix
